

**DESCRIPTION**

APZ 3420 x is a flame proof version of our standard industrial pressure transmitter. Reliable piezoresistive sensor technology combined with fully welded stainless steel case offer a versatile yet cost-effective transmitter in Ex d compact form factor. Optionally available with a discrete output signal (PNP) in addition to the analog signal.

SPECIFICATIONS

Pressure ranges: 0...40 mbar to 0...600 bar

Pressure types: gauge, absolute

Basic accuracy: $\pm 0.25\%$

Outputs: 4...20 mA; 4...20 mA / PNP; 0...20 mA; 0...10 B; 0...5 B; HART®; RS-485 / Modbus RTU

Sensor: silicon piezoresistive

Pressure port: G1/2"; G1/4"; 1/2" NPT; 1/4" NPT; M20x1,5; other

Media temperature: -40...+125 °C

Ambient temperature: -40...+85 °C

Optional graphics display with relay output (PNP)

Explosion proof: 1Ex d IIC T6...T4 Gb X

HART® is a registered trademark of the HART Communication Foundation.

APPLICATIONS

Oil field equipment

Upstream oil and gas production

Gas compressors

Oil platforms

TECHNICAL SPECIFICATIONS

MEASURING RANGES

Pressure range, bar		Overpressure, bar	Burst pressure, bar	Диапазон давления, бар		Overpressure, bar	Burst pressure, bar
Gauge	Absolute			Gauge	Absolute		
-1...0	-	3	4	0...16	0...16	60	80
0...0.4	0...0.4	1	1.5	0...25	0...25	60	80
0...0.6	0...0.6	3	4	0...40	0...40	100	150
0...1	0...1	3	4	0...60	0...60	100	150
0...1.6	0...1.6	6	8	0...100	0...100	150	230
0...2.5	0...2.5	6	8	0...160	0...160	300	450
0...4	0...4	15	15	0...250	0...250	530	780
0...6	0...6	15	20	0...400	0...400	1050	1580
0...10	0...10	30	40	0...600	0...600	1050	1580

PERFORMANCE	P > 0.4 bar	0.4 ≥ P > 0.16 bar
Accuracy, % of span*	≤ ±0.25 (standard) / 0.20 (optional)	≤ ±0.5
Temperature effect (% of span / 10 °C)	≤ ±0.1	≤ ±0.2
Compensated range	-20...+80 °C	0...+80 °C
Compensated range (optional)	-40...+60 °C	-40...+60 °C
Power supply effect	≤ ±0.05% of span / 10 V	
Load resistance effect	≤ ±0.05% of span / kOhm (transmitters with current output)	
Long-term stability	≤ ±0.1% of span / year	
Response time (10...90%)	≤ 1 ms with analog output, ≤ 200 ms with digital output	

* Accuracy includes non-linearity, hysteresis and non-repeatability.

OPERATING CONDITIONS

Medium temperature (depends on seal)	-40...+125 °C		
Ambient temperature	-40...+85 °C		
Storage temperature	-50...+85 °C		
Approvals	1Ex d IIC T6...T4 Gb X		
Temperature class	T4	T5	T6
Ambient temperature	-40...85 °C	-40...70 °C	-40...60 °C
Vibration resistance	10 g RMS, 20–2000 Hz		
Shock resistance	100 g / 11 ms		
Service life	> 100 x 10 ⁶ cycles		

MECHANICAL SPECIFICATIONS

Pressure port material	stainless steel 304 (1.4301) – standard; stainless steel 316L (1.4404) – optional		
Housing material	stainless steel 316L (1.4404)		
Seal	welded (-40...+125 °C)		
Diaphragm	stainless steel 316L (1.4435)		
Wetted parts	Diaphragm, pressure port, seal		
Pressure port	M20x1,5 EN 837; G1/2" EN 837; G1/4" EN 837; M12x1 EN 837; M12x1,5 EN 837; 1/4" – 18 NPT; 1/2" – 14 NPT; G1/4" DIN 3852; other		
Electrical connection	Ingress protection	Cross-section	Cable diameter
Field housing, cable gland M20x1.5	IP67	1.5 mm ²	6...12 mm

DIGITAL DISPLAY

Display type	OLED 128x64 pixels (30x16 mm)
Displayed units	bar, mbar, MPa, kPa, Pa, psi, mmHg, mH ₂ O, ftH ₂ O, %, mA, user
Displayed values range	-1999...9999
Display accuracy	0.1 % of span ± 1 digit
Settling time	< 1 s (with damping disabled)
Damping	0.3...30 s (programmable)

ELECTRICAL SPECIFICATIONS

Output signal	Power supply, U_s	Load resistance, R	Power consumption
4...20 mA / 2-wire	12...36 V	$\leq [(U_s - 12 \text{ V}) / 0.02 \text{ A}] \text{ Ohm}^*$	$\leq 26 \text{ mA}$
4...20 mA / HART®	18...42 V (with display)	$\leq [(U_s - 18 \text{ V}) / 0.02 \text{ A}] \text{ Ohm}^*$ (with display)	
4...20 mA / 3-wire	12...36 V	$\leq 500 \text{ Ohm}$	
0...20 mA / 3-wire		$\geq 10 \text{ kOhm}$	$< 7 \text{ mA}$
0...10 V / 3-wire		$\geq 5 \text{ kOhm}$	
0...5 V / 3-wire			
0.5...4.5 V / 3-wire	5 V	$\leq 5 \text{ kOhm}$	$\leq 2 \text{ mA}$
0.5...4.5 V / 3-wire	6...15 V		$\leq 7 \text{ mA}$
RS-485 / Modbus RTU	12...36 V	-	

* For output signal 4...20 mA / HART® minimum load resistance for digital communication: 250 Ohm.

DISCRETE OUTPUT TYPE

Discrete output type	PNP
Max switching voltage	36 V (DC)
Switching current (DC)	$\leq 30 \text{ mA}$
Voltage drop	$\leq 1.5 \text{ V}$
Operating modes (locally configurable)	Hysteresis / Window
Switching delay (locally configurable)	0...100 s
Repeatability	$\leq 0.1\%$ of span
Accuracy	$\leq 0.5\%$ of span
Maximum switching frequency	10 Hz

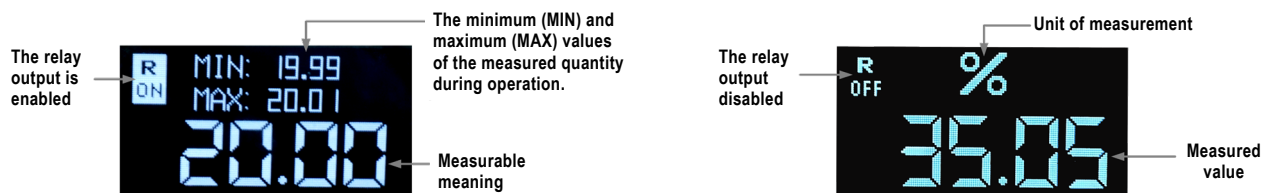
ELECTRICAL CONNECTIONS / PIN ASSIGNMENT

Circuits	Field housing without display	Field housing with display	With display and PNP discrete output
2-wire	power +	2	2
	power -	3	3
	shield	1	1
	PNP	-	4
3-wire	power +	2	-
	power -	3	-
	signal +	4	-
	shield	1	-
RS-485 4-wire	power +	2	-
	power -	3	-
	A	1	-
	B	4	-
	shield	Housing	-

THE FRONT PANEL ELEMENTS

The front panel of the device features two control buttons:  (forward) and  (back), which, when the device is in operation, toggle between the types of information displayed on the screen (numeric or graphical).

To enter the parameter setting mode, briefly press both buttons  +  – simultaneously; the same action is required to save the set value. Holding both buttons down simultaneously returns to the previous screen or exits the setting mode and returns to the operating mode to display the measured parameters.

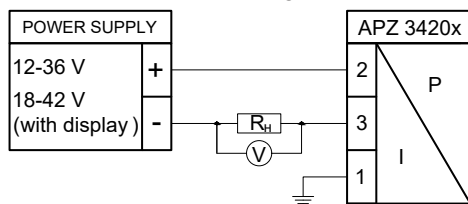


Examples of the device's switchable screens

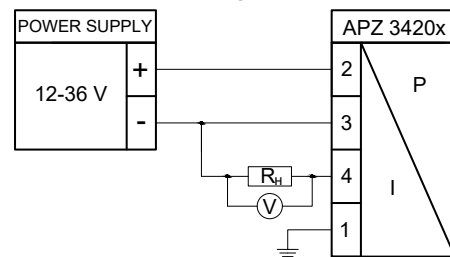
The display can be rotated in 90° increments.

APPLICATION CONNECTIONS

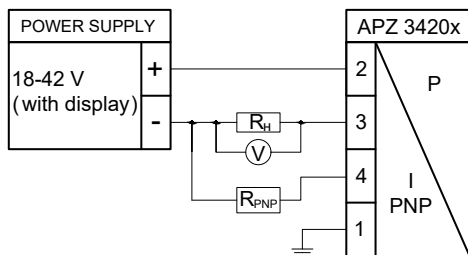
2-wire



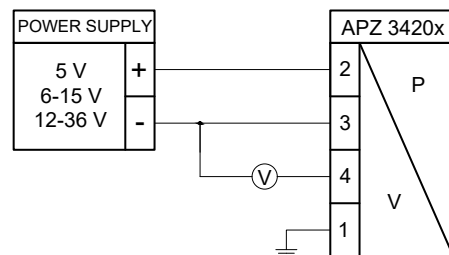
3-wire (output signal - current)



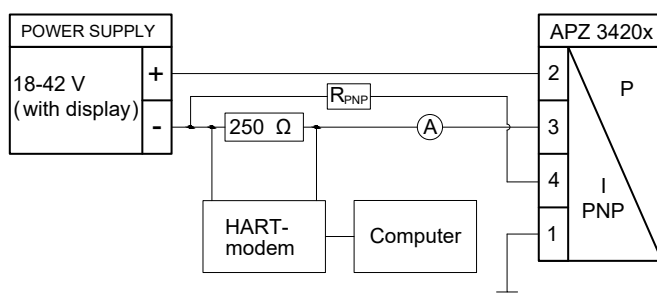
2-wire (with discrete output signal)



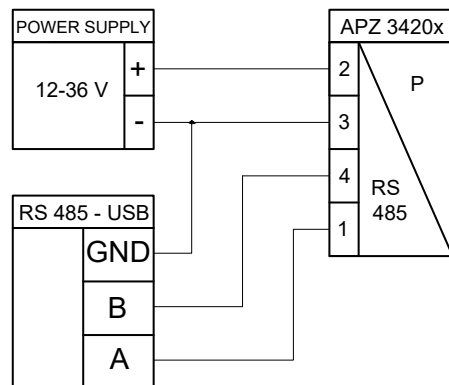
3-wire (output signal voltage)



2-wire (with discrete output signal + HART)

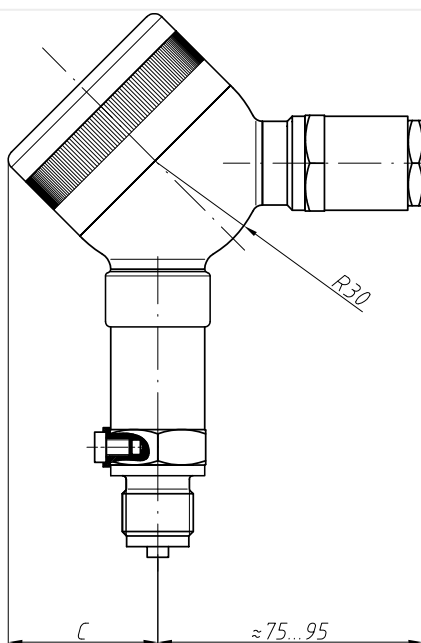


4-wire (output signal - RS 485)

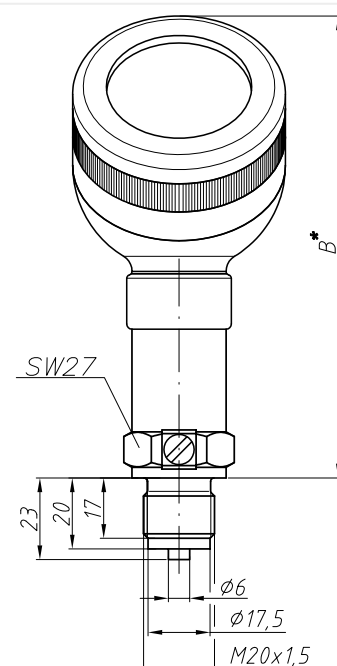


DIMENSIONS (mm)

Field housing (with M20x1,5 EN 837)

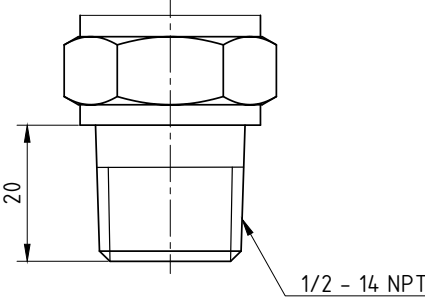
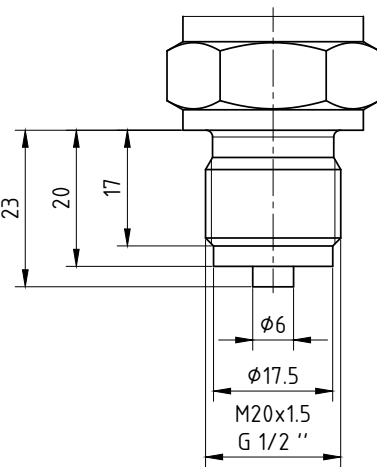
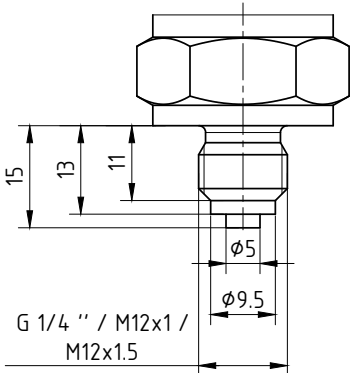
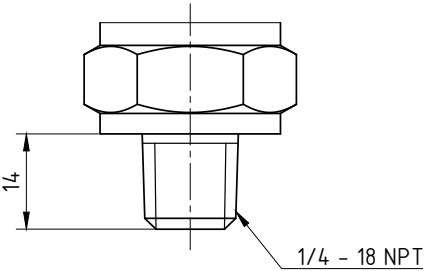
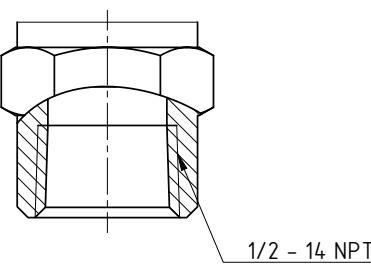
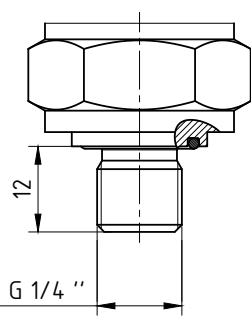


	B	C
with display	130	42
without display	127	39



* Housing of RS-485 / Modbus RTU with is 25 mm longer
Housing of HART® version is 30 mm longer

PRESSURE PORTS, DIMENSIONS (mm)

1/2" - 14 NPT; code 820	M20x1,5 EN 837; code 201 G1/2" EN 837; code 721	G1/4" EN 837; code 741 M12x1 EN 837; code 121 M12x1,5 EN 837; code 123
 Technical drawing of a pressure port with a hexagonal base and a cylindrical body. The total height is 20 mm. The bottom thread is labeled 1/2 - 14 NPT.	 Technical drawing of a pressure port with a hexagonal base and a cylindrical body. The total height is 23 mm. The body has a height of 20 mm and a base diameter of 17 mm. The bottom thread is labeled M20x1,5 G 1/2 ". The internal thread is labeled 1/2 - 14 NPT.	 Technical drawing of a pressure port with a hexagonal base and a cylindrical body. The total height is 15 mm. The body has a height of 13 mm and a base diameter of 11 mm. The bottom thread is labeled G 1/4 " / M12x1 / M12x1,5. The internal thread is labeled 1/2 - 14 NPT.
1/4" - 18 NPT; code 840	1/2" - 14 NPT internal thread; code 821	G1/4" DIN 3852; code 740
 Technical drawing of a pressure port with a hexagonal base and a cylindrical body. The total height is 14 mm. The bottom thread is labeled 1/4 - 18 NPT.	 Technical drawing of a pressure port with a hexagonal base and a cylindrical body. The total height is 14 mm. The bottom thread is labeled 1/2 - 14 NPT.	 Technical drawing of a pressure port with a hexagonal base and a cylindrical body. The total height is 12 mm. The bottom thread is labeled G 1/4 ".

ORDERING CODE

APZ 3420 x		-X	-X	-XXXX	-X	-XX	-XX	-XXX	-X	-XX
EXPLOSION-PROOF ENCLOSURE										
1Ex d IIC T6...T4 Gb X										
MEASUREMENT TYPE										
Gauge		G								
Absolute		A								
Vacuum, LRL = -1 bar		V								
UNIT OF MEASUREMENT										
bar		B								
kg/cm²		S								
mH ₂ O		W								
kPa		H								
MPa		M								
Other (specify when ordering)		X								
UPPER RANGE LIMIT (URL)										
bar, kg/cm²	mH ₂ O	kPa	MPa							
0.4	0400	4	4000	40	4001					
0.6	0600	6	6000	60	6001					
1	1000	10	1001	100	1002	0.1	0100			
1.6	1600	16	1601	160	1602	0.16	0160			
2.5	2500	25	2501	250	2502	0.25	0250			
4	4000	40	4001	400	4002	0.4	0400			
6	6000	60	6001	600	6002	0.6	0600			
10	1001	100	1002	1000	1003	1	1000			
16	1601	160	1602			1.6	1600			
25	2501	250	2502			2.5	2500			
40	4001	400	4002			4	4000			
60	6001					6	6000			
100	1002					10	1001			
160	1602					16	1601			
250	2502					25	2501			
400	4002					40	4001			
600	6002					60	6001			
Other	XXXX	Other	XXXX	Other	XXXX	Other	XXXX			
2 – RANGE		XXXX-XXXX								
3 – RANGE		XXXX-XXXX-XXXX								
ACCURACY										
0.2% (P > 0.4 bar)		B								
0.25% (P > 0.4 bar) (standard)		C								
0.5%		D								
Other (specify when ordering)		X								
ELECTRICAL CONNECTION										
Field housing without display, Exd cable entry M20x1.5 for armored cable 6-12 mm, armor 9-17 mm		62								
Field housing without display, Exd cable entry M20x1.5 for unarmored cable 6-12 mm, in metal hose 15 mm		63								
Field housing with display, Exia/Exd cable entry M20x1.5 for armored cable 6-12 mm, armor 9-17 mm		68								
Field housing with display, Exia/Exd cable entry M20x1.5 for unarmored cable 6-12 mm, in metal hose 15 mm		69								
Other (specify when ordering)		XX								

ORDERING CODE (CONTINUED)

	APZ 3420 x	-X	-X	-XXXX	-X	-XX	-XX	-XXX	-X	-XX
OUTPUT SIGNAL										
							4...20 mA / 2-wire (standard)	A		
							4...20 mA / 2-wire / PNP (with display only)	AP		
							4...20 mA / 3-wire	B		
							0...20 mA / 3-wire	C		
							0...5 mA / 3-wire	S		
							0...10 V / 3-wire	D		
							0...5 V / 3-wire	E		
							0.5...4.5 V / 3-wire, supply 5 V	R2		
							0.5...4.5 V / 3-wire, supply 6...15 V	K		
							RS-485 / Modbus RTU	M		
							4...20 mA / HART®	H		
							4...20 mA / HART / PNP (only with display)	HP		
							Other (specify when ordering)	X		
PRESSURE PORT										
							M20x1.5 EN 837	201		
							G1/2" EN 837	721		
							G1/4" EN 837	741		
							M12x1 EN 837	121		
							M12x1.5 EN 837	123		
							1/2" – 14 NPT internal	821		
							1/4" – 18 NPT	840		
							1/2" – 14 NPT	820		
							G1/4" DIN 3852	740		
							Other (specify when ordering)	XXX		
SEALS										
							Welded (-40...+125 °C)	W		
							Other (specify when ordering)	X		
VERSION										
							Standard	00		
							Temperature compensated in the range of -40...+60 °C	46		
							Compound filled version	16		
							Other (specify when ordering)	XX		

Example: APZ 3420 x-G-B-4001-B-62-A-201-W-00

ACCESSORIES

				
DZ 10 Pressure snubber	PZ 1024 Power supply unit	BZ 05 / BZ 10 Dry air junction box for submersible transmitters		